

AYMANE AIT DADS

Embedded AI Intern | PyTorch · FP16 · Hardware-Aware Optimization

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Hiring Team - Embedded AI Intern [IDA:00027]

Aumovio · Frankfurt, Germany

Dear Aumovio Hiring Team,

Your requirement to **evaluate and optimize AI models under embedded constraints** - latency, memory, and accuracy - maps directly to engineering decisions I have already made under real hardware pressure. Fine-tuning **CLIP ViT-L/14 (428M parameters)** on a single T4 GPU, I applied **FP16 mixed precision with GradScaler** to stay within memory bounds while preserving inference accuracy, and ranked #1 on the EURECOM private leaderboard across 250K training samples.

On the evaluation and benchmarking side, I ran structured **ablation experiments** across transformer block unfreezing depth (4, 6, and 8 layers), learning rate schedules, and augmentation strategies - exactly the kind of **trade-off analysis between optimization approaches** your role requires. Separately, my **transformer autoencoder** for unsupervised industrial audio fault detection reached **AUC 0.80** with no labeled anomalies, demonstrating that I design for deployability under constrained, real-world conditions rather than clean benchmark settings.

Aumovio sits at the intersection of automotive safety and software-defined vehicle architecture - a context where an AI model that fails silently at inference time on an MCU is not an academic problem but a safety one. I want to contribute to that constraint directly. I would welcome the chance to discuss how my optimization and evaluation work fits the team's current embedded targets.

Sincerely,

AYMANE AIT DADS